

CHAPTER 8: INDEX MODELS

5. a. To optimize this portfolio one would need:

$n = 60$ estimates of means

$n = 60$ estimates of variances

$$\frac{n^2 - n}{2} = 1,770 \text{ estimates of covariances}$$

Therefore, in total: $\frac{n^2 + 3n}{2} = 1,890$ estimates

- b. In a single index model: $r_i - r_f = \alpha_i + \beta_i(r_M - r_f) + e_i$

Equivalently, using excess returns: $R_i = \alpha_i + \beta_i R_M + e_i$

The variance of the rate of return can be decomposed into the components:

(1) The variance due to the common market factor: $\beta_i^2 \sigma_M^2$

(2) The variance due to firm specific unanticipated events: $\sigma^2(e_i)$

In this model: $\text{Cov}(r_i, r_j) = \beta_i \beta_j \sigma$

The number of parameter estimates is:

$n = 60$ estimates of the mean $E(r_i)$

$n = 60$ estimates of the sensitivity coefficient β_i

$n = 60$ estimates of the firm-specific variance $\sigma^2(e_i)$

1 estimate of the market mean $E(r_M)$

1 estimate of the market variance σ_M^2

Therefore, in total, 182 estimates.

The single index model reduces the total number of required estimates from 1,890 to 182. In general, the number of parameter estimates is reduced from:

$$\left(\frac{n^2 + 3n}{2} \right) \text{ to } (3n + 2)$$

6. a. The standard deviation of each individual stock is given by:

$$\sigma_i = [\beta_i^2 \sigma_M^2 + \sigma^2(e_i)]^{1/2}$$

Since $\beta_A = 0.8$, $\beta_B = 1.2$, $\sigma(e_A) = 30\%$, $\sigma(e_B) = 40\%$, and $\sigma_M = 22\%$, we get:

$$\sigma_A = (0.8^2 \times 22^2 + 30^2)^{1/2} = 34.78\%$$

$$\sigma_B = (1.2^2 \times 22^2 + 40^2)^{1/2} = 47.93\%$$

- b. The expected rate of return on a portfolio is the weighted average of the expected returns of the individual securities:

$$E(r_P) = w_A \times E(r_A) + w_B \times E(r_B) + w_f \times r_f$$

$$E(r_P) = (0.30 \times 13\%) + (0.45 \times 18\%) + (0.25 \times 8\%) = 14\%$$

The beta of a portfolio is similarly a weighted average of the betas of the individual securities:

$$\beta_P = w_A \times \beta_A + w_B \times \beta_B + w_f \times \beta_f$$

$$\beta_P = (0.30 \times 0.8) + (0.45 \times 1.2) + (0.25 \times 0.0) = 0.78$$

The variance of this portfolio is:

$$\sigma_P^2 = \beta_P^2 \sigma_M^2 + \sigma^2(e_P)$$

where $\beta_P^2 \sigma_M^2$ is the systematic component and $\sigma^2(e_P)$ is the nonsystematic component. Since the residuals (e_i) are uncorrelated, the nonsystematic variance is:

$$\sigma^2(e_P) = w_A^2 \times \sigma^2(e_A) + w_B^2 \times \sigma^2(e_B) + w_f^2 \times \sigma^2(e_f)$$

$$= (0.30^2 \times 30^2) + (0.45^2 \times 40^2) + (0.25^2 \times 0) = 405$$

where $\sigma^2(e_A)$ and $\sigma^2(e_B)$ are the firm-specific (nonsystematic) variances of Stocks A and B, and $\sigma^2(e_f)$, the nonsystematic variance of T-bills, is zero. The residual standard deviation of the portfolio is thus:

$$\sigma(e_P) = (405)^{1/2} = 20.12\%$$

The total variance of the portfolio is then:

$$\sigma_P^2 = (0.78^2 \times 22^2) + 405 = 699.47$$

The total standard deviation is 26.45%.

7. a. The two figures depict the stocks' security characteristic lines (SCL). Stock A has higher firm-specific risk because the deviations of the observations from the SCL are larger for Stock A than for Stock B. Deviations are measured by the vertical distance of each observation from the SCL.
- b. Beta is the slope of the SCL, which is the measure of systematic risk. The SCL for Stock B is steeper; hence Stock B's systematic risk is greater.

- c. The R^2 (or squared correlation coefficient) of the SCL is the ratio of the explained variance of the stock's return to total variance, and the total variance is the sum of the explained variance plus the unexplained variance (the stock's residual variance):

$$R^2 = \frac{\beta_i^2 \sigma_M^2}{\beta_i^2 \sigma_M^2 + \sigma^2(e_i)}$$

Since the explained variance for Stock B is greater than for Stock A (the explained variance is $\beta_B^2 \sigma_M^2$, which is greater since its beta is higher), *and* its residual variance $\sigma^2(e_B)$ is smaller, its R^2 is higher than Stock A's.

- d. Alpha is the intercept of the SCL with the expected return axis. Stock A has a small positive alpha whereas Stock B has a negative alpha; hence, Stock A's alpha is larger.
- e. The correlation coefficient is simply the square root of R^2 , so Stock B's correlation with the market is higher.

8. a. Firm-specific risk is measured by the residual standard deviation. Thus, stock A has more firm-specific risk: $10.3\% > 9.1\%$
- b. Market risk is measured by beta, the slope coefficient of the regression. A has a larger beta coefficient: $1.2 > 0.8$
- c. R^2 measures the fraction of total variance of return explained by the market return. A's R^2 is larger than B's: $0.576 > 0.436$
- d. Rewriting the SCL equation in terms of total return (r) rather than excess return (R):

$$r_A - r_f = \alpha + \beta \times (r_M - r_f) \Rightarrow$$

$$r_A = \alpha + r_f \times (1 - \beta) + \beta \times r_M$$

The intercept is now equal to:

$$\alpha + r_f \times (1 - \beta) = 1\% + r_f \times (1 - 1.2)$$

Since $r_f = 6\%$, the intercept would be: $1\% + 6\%(1 - 1.2) = 1\% - 1.2\% = -0.2\%$

9. The standard deviation of each stock can be derived from the following equation for R^2 :

$$R_i^2 = \frac{\beta_i^2 \sigma_M^2}{\sigma_i^2} = \frac{\text{Explained variance}}{\text{Total variance}}$$

Therefore:

$$\sigma_A^2 = \frac{\beta_A^2 \sigma_M^2}{R_A^2} = \frac{0.7^2 \times .20^2}{0.20} = .098$$

$$\sigma_A = .3130 = 31.30\%$$

For stock B:

$$\sigma_B^2 = \frac{\beta_B^2 \sigma_M^2}{R_B^2} = \frac{1.2^2 \times .20^2}{0.12} = .048$$

$$\sigma_B = .6928 = 69.28\%$$

10. The systematic risk for A is:

$$\beta_A^2 \times \sigma_M^2 = 0.70^2 \times 20^2 = 196$$

The firm-specific risk of A (the residual variance) is the difference between A's total risk and its systematic risk:

$$980 - 196 = 784$$

The systematic risk for B is:

$$\beta_B^2 \times \sigma_M^2 = 1.20^2 \times 20^2 = 576$$

B's firm-specific risk (residual variance) is:

$$4,800 - 576 = 4,224$$

11. The covariance between the returns of A and B is (since the residuals are assumed to be uncorrelated):

$$\text{Cov}(r_A, r_B) = \beta_A \beta_B \sigma_M^2 = 0.70 \times 1.20 \times .04 = .0336$$

The correlation coefficient between the returns of A and B is:

$$\rho_{AB} = \frac{\text{Cov}(r_A, r_B)}{\sigma_A \sigma_B} = \frac{.0336}{.3130 \times .6928} = .1549$$

12. Note that the correlation is the square root of R^2 : $\rho = \sqrt{R^2}$

$$\text{Cov}(r_A, r_M) = \rho \sigma_A \sigma_M = 0.20^{1/2} \times 31.30 \times 20 = 280$$

$$\text{Cov}(r_B, r_M) = \rho \sigma_B \sigma_M = 0.12^{1/2} \times 69.28 \times 20 = 480$$

13. For portfolio P we can compute:

$$\sigma_P = [(0.6^2 \times 980) + (0.4^2 \times 4800) + (2 \times 0.4 \times 0.6 \times 336)]^{1/2} = [1282.08]^{1/2} = 35.81\%$$

$$\beta_P = (0.6 \times 0.7) + (0.4 \times 1.2) = 0.90$$

$$\sigma^2(e_P) = \sigma_P^2 - \beta_P^2 \sigma_M^2 = 1282.08 - (0.90^2 \times 400) = 958.08$$

$$\text{Cov}(r_P, r_M) = \beta_P \sigma_M^2 = 0.90 \times 400 = 360$$

This same result can also be attained using the covariances of the individual stocks with the market:

$$\begin{aligned} \text{Cov}(r_P, r_M) &= \text{Cov}(0.6r_A + 0.4r_B, r_M) = 0.6 \times \text{Cov}(r_A, r_M) + 0.4 \times \text{Cov}(r_B, r_M) \\ &= (0.6 \times 280) + (0.4 \times 480) = 360 \end{aligned}$$

14. Note that the variance of T-bills is zero, and the covariance of T-bills with any asset is zero. Therefore, for portfolio Q :

$$\begin{aligned} \sigma_Q &= [w_P^2 \sigma_P^2 + w_M^2 \sigma_M^2 + 2 \times w_P \times w_M \times \text{Cov}(r_P, r_M)]^{1/2} \\ &= [(0.5^2 \times 1,282.08) + (0.3^2 \times 400) + (2 \times 0.5 \times 0.3 \times 360)]^{1/2} = 21.55\% \end{aligned}$$

$$\beta_Q = w_P \beta_P + w_M \beta_M = (0.5 \times 0.90) + (0.3 \times 1) + (0.20 \times 0) = 0.75$$

$$\sigma^2(e_Q) = \sigma_Q^2 - \beta_Q^2 \sigma_M^2 = 464.52 - (0.75^2 \times 400) = 239.52$$

$$\text{Cov}(r_Q, r_M) = \beta_Q \sigma_M^2 = 0.75 \times 400 = 300$$

15. a. Beta Books adjusts beta by taking the sample estimate of beta and averaging it with 1.0, using the weights of 2/3 and 1/3, as follows:

$$\text{adjusted beta} = [(2/3) \times 1.24] + [(1/3) \times 1.0] = 1.16$$

- b. If you use your current estimate of beta to be $\beta_{t-1} = 1.24$, then

$$\beta_t = 0.3 + (0.7 \times 1.24) = 1.168$$

16. For Stock A:

$$\alpha_A = r_A - [r_f + \beta_A \times (r_M - r_f)] = .11 - [.06 + 0.8 \times (.12 - .06)] = 0.2\%$$

For stock B:

$$\alpha_B = r_B - [r_f + \beta_B \times (r_M - r_f)] = .14 - [.06 + 1.5 \times (.12 - .06)] = -1\%$$

Stock A would be a good addition to a well-diversified portfolio. A short position in Stock B may be desirable.

CFA PROBLEMS

1. The regression results provide quantitative measures of return and risk based on monthly returns over the five-year period.

β for ABC was 0.60, considerably less than the average stock's β of 1.0. This indicates that, when the S&P 500 rose or fell by 1 percentage point, ABC's return on average rose or fell by only 0.60 percentage point. Therefore, ABC's systematic risk (or market risk) was low relative to the typical value for stocks. ABC's alpha (the intercept of the regression) was -3.2% , indicating that when the market return was 0%, the average return on ABC was -3.2% . ABC's unsystematic risk (or residual risk), as measured by $\sigma(e)$, was 13.02%. For ABC, R^2 was 0.35, indicating closeness of fit to the linear regression greater than the value for a typical stock.

β for XYZ was somewhat higher, at 0.97, indicating XYZ's return pattern was very similar to the β for the market index. Therefore, XYZ stock had average systematic risk for the period examined. Alpha for XYZ was positive and quite large, indicating a return of 7.3%, on average, for XYZ independent of market return. Residual risk was 21.45%, half again as much as ABC's, indicating a wider scatter of observations around the regression line for XYZ. Correspondingly, the fit of the regression model was considerably less than that of ABC, consistent with an R^2 of only 0.17.

The effects of including one or the other of these stocks in a diversified portfolio may be quite different. If it can be assumed that both stocks' betas will remain stable over time, then there is a large difference in systematic risk level. The betas obtained from the two brokerage houses may help the analyst draw inferences for the future. The three estimates of ABC's β are similar, regardless of the sample period of the underlying data. The range of these estimates is 0.60 to 0.71, well below the market average β of 1.0. The three estimates of XYZ's β vary significantly among the three sources, ranging as high as 1.45 for the weekly data over the most recent two years. One could infer that XYZ's β for the future might be well above 1.0, meaning it might have somewhat greater systematic risk than was implied by the monthly regression for the five-year period.

These stocks appear to have significantly different systematic risk characteristics. If these stocks are added to a diversified portfolio, XYZ will add more to total volatility.

2. The R^2 of the regression is: $0.70^2 = 0.49$

Therefore, 51% of total variance is unexplained by the market; this is nonsystematic risk.

3. $9 = 3 + \beta (11 - 3) \Rightarrow \beta = 0.75$

4. d.

5. b.